

Food Insecurity Forecast Charts - Script Instructions

How to run the Python script that turns a food insecurity forecast CSV into review-ready charts, summary tables, and an HTML dashboard.

The chart script is designed for forecast files with one row per ZIP/ZCTA-county combination. It helps reviewers quickly see which communities have the highest predicted rates, where the largest estimated food-insecure populations are, and which input drivers move with the forecast.

Key idea. The input forecast CSV is required at runtime. This keeps the script reusable across forecast years and avoids accidentally charting an older file.

Quick Start RUN COMMAND

From the project root, run:

```
python3 scripts/create_food_insecurity_forecast_charts.py --input forecasts/fl_food_insecurity_forecast_2027.csv
```

To choose a specific output folder, add `--output-dir`:

```
python3 scripts/create_food_insecurity_forecast_charts.py --input forecasts/fl_food_insecurity_forecast_2027.csv --output-dir reports/figures/fl_food_insecurity_forecast_2027
```

The script also accepts `--top-n`, which controls how many ZIP/ZCTA or county rows appear in ranked bar charts. The default is 25.

Inputs WHAT THE CSV NEEDS

COLUMN	PURPOSE
zcta	ZIP Code Tabulation Area identifier shown in ZIP/ZCTA charts.
County, State	County label used for county summaries and ZIP/ZCTA chart labels.
population	Used to estimate food-insecure people and calculate weighted rates.
predicted_food_insecurity_rate	Main model prediction used in every rate chart.
forecast_year	Used in the dashboard title when present.
Food Bank 1	Used for the food-bank boxplot and food-bank summary when present.
poverty_rate and other numeric drivers	Used in the scatterplot color scale and driver correlation chart when present.

Rows with missing predicted rates are left out of rate-based charts and summaries. They remain in the source CSV; the charting step simply avoids plotting blank predictions.

Outputs WHAT GETS CREATED

OUTPUT	HOW TO USE IT
forecast_dashboard.html	Open this first. It pulls the key metrics, county table, food-bank table, and all generated chart images into one review page.
top_zctas_by_rate.png	Shows the ZIP/ZCTA-county rows with the highest predicted food insecurity rates.
top_zctas_by_estimated_people.png	Highlights the ZIP/ZCTA-county rows with the largest estimated number of food-insecure people.
county_weighted_rates.png	Ranks counties by population-weighted predicted food insecurity rate.

OUTPUT	HOW TO USE IT
county_estimated_people.png	Ranks counties by estimated number of food-insecure people.
rate_distribution.png	Shows whether predictions are clustered tightly or spread across a wide range.
population_vs_predicted_rate.png	Compares ZIP/ZCTA population with predicted rate. Poverty rate is used as color when available.
food_bank_rate_boxplot.png	Compares the range of predicted rates across food-bank service areas.
driver_correlations.png	Shows which numeric variables are most correlated with the predicted rate in this forecast file.
county_summary.csv, food_bank_summary.csv, top_zctas*.csv	Reusable summary tables for spreadsheets, slides, or follow-up analysis.

How to Read the Charts INTERPRETATION GUIDE

CHART	WHAT TO LOOK FOR
Highest predicted rates	Places where the modeled share of residents facing food insecurity is highest. These are rate hot spots, not necessarily the largest populations.
Largest estimated people	Places where predicted rate multiplied by population is largest. These are useful for scale and planning volume.
County weighted rates	County-level rates weighted by ZIP/ZCTA population, which gives large and small ZIP/ZCTAs proportional influence.
Distribution	The overall spread of forecasts. A long right tail indicates a smaller set of ZIP/ZCTAs with notably higher predicted rates.
Population scatterplot	Whether high-rate places are also large-population places. Points far above the main cloud deserve attention.

CHART	WHAT TO LOOK FOR
Driver correlations	A descriptive screening view. Correlation does not prove a driver caused the prediction, but it helps explain what varies with the forecast.

Common Issues TROUBLESHOOTING

If the script says a required column is missing: confirm the input file is the forecast output, not a raw MMG workbook or a source-data extract. The chart script expects a scored forecast CSV with `predicted_food_insecurity_rate`.

If charts are not where expected: check the printed output directory at the end of the run. Without `--output-dir`, the default folder is `reports/figures/fl_food_insecurity_forecast_2027`.

If the script will not start: make sure the command is run from the repository root and that Python can import `pandas`, `numpy`, and `matplotlib`.